LANG'S MPSS · MULTI-PURPOSE SEMI-SUBMERSIBLE

DRAFT · FOR REVIEW BEFORE RELEASE

Founding Investor Memorandum

Turning the offshore production host from a bespoke project into leaseable infrastructure.

> Seaways MPSS is a validated, independently classed production host built with proven ship-building technology. This memorandum sets out the opportunity for founding investors: to capitalise the platform that converts a four-decade, class-endorsed design into a fleet of standardised, income-producing hulls leased to offshore operators across the field envelope — from marginal fields in shallow water through to deepwater.

SIMPLE · SAFE · SWIFT · SIZE · SAVES

PREPARED FOR
Founding investors & strategic partners

ASSET CLASS
Real assets · offshore infrastructure

INSTRUMENT
Founding equity (illustrative)

STATUS
Draft · for discussion only

A product where the industry only has projects.

Every offshore development engineers a new floating host from scratch — a multi-year, multi-billion dollar bottleneck that sits directly between a discovery and its first barrel. The MPSS replaces that bespoke project with a validated standard host: one classed hull, adapted by its topside package, working the full field envelope from marginal shallow-water developments to deepwater, and available to lease before final investment decision. It is the difference between building a project and owning a product.

15,000 t	**~14** mo	**LR·ABS** DNV	**40** yrs
Deck payload on a large, stable, open deck	Estimated standardised hull delivery	Independently classed & endorsed	Of engineering, tank-testing & review

1. **The pain is universal and expensive**

   Between host selection and first production, operators carry years of schedule risk and billions in committed capital before host risk is even settled.

2. **The design already exists and is validated**

   The five-S MPSS has been analysed, tank-tested, peer-reviewed and reviewed for classification by Lloyd's Register, ABS and DNV — the risky R&D is behind it, not ahead of it.

3. **The business model is leasing, not selling**

   A standardised hull is leaseable inventory. Multi-year day-rate charters throw off contracted, uncorrelated cash flow; the redeployable hull retains hard-asset residual value.

4. **Founding capital converts design into fleet**

   This round funds the first standardised host to a shipyard-ready package and stands up the leasing platform, Lang Leasing — the step that turns a validated drawing into a contracted, earning asset.

Deepwater value is won or lost between host selection and first oil.

Every offshore project runs the same gauntlet. The stretch from **FID to First Production** — Select, Define/FEED, FID, Execute — is the "kill zone" where operators bleed schedule and capital. The root cause is that the host itself is re-invented every single time.

THE CONVENTIONAL PATH

- A new host engineered around every field
- Long, custom FEED & risk-reduction cycle
- Heavy long-lead procurement
- Bespoke, engineering-intensive fabrication
- Billions committed at FID before host risk is settled
- Revenue delayed by host delivery

THE MPSS PATH

- A validated standard host — classed and known
- Known hull, known motions, known fabrication
- The host becomes leaseable inventory before FID
- Repeatable, predictable build
- Lease structure reduces upfront capital exposure
- Faster availability accelerates first oil

The conventional model designs a new host around every field. The MPSS reverses it — standard host first, field-specific topsides second. That is the difference between a product and a project.

Not new technology — proven technology, applied properly.

The Multi-Purpose Semi-Submersible uses standard ship-building methods: uniform box construction in stiffened mild-steel plate — a continuous ring pontoon, box columns and a buoyant deck box with right-angle joints that are simple to weld and easy to inspect. Less complex than conventional semis, with excellent structural continuity, and adapted to any mission by its topside package.

S **Simple**

Modular, panel-line fabrication in common shipyard practice. Standard mild steel, a single operating draft, simple ballast and trim.

S **Safe**

A massive reserve of buoyancy and watertight subdivision. Tested to a 37.5 m (123 ft) storm wave — no water on deck, broken-mooring survival.

S **Swift**

A modular design means a hull in about 14 months; prefabricated topsides skid on before deck assembly for fast turnaround.

S **Size**

In the ocean, bigger is better — more stable, more deck, more storage, reconfigurable. Ample margin for steel catenary risers and dry trees.

S **Saves**

Low initial cost, low maintenance, low operating cost through standardised box construction — plus high residual value.

HULL ATTRIBUTE	CONVENTIONAL TWIN-PONTOON FPU	LANG'S MPSS
Pontoon form	Twin ship-form pontoons	Continuous ring pontoon
Construction	Bespoke, engineering-intensive	Standardised mild-steel box, panel-line built
Operating draft	Changes to a survival draft in heavy seas	Single deep operating draft — constant production
Deck & motion	Truss-supported, reacts to surface waves	Large open deck, low heave from deep-draft damping
Onboard storage	Typically none — needs an FPSO or tanker	~300,000 bbl dry above the waterline, plus up to 2 M bbl oil-over-water in the "Big SWIS" lower tank

A standard host meets a market that keeps re-building bespoke ones.

Offshore fields — from marginal developments in shallow water through to deepwater — remain one of the largest sources of new barrels, and every project still commissions a one-off host. The same standardised, low-motion hull also answers a widening set of offshore missions beyond oil & gas — which is where a leaseable platform compounds.

Marginal-to-deepwater production & tiebacks

The core market: a low-motion host that works the full depth envelope, strengthens the SCR / riser / cable case and carries 15,000 t of topsides on an open deck.

Storage & FPSO-class duty

Around 300,000 bbl of dry storage above the waterline, and up to 2 M bbl of oil-over-water in the large lower tank ("Big SWIS") — FPSO-class capacity without a bespoke conversion.

The energy transition

Second and third lives in power, compression, CCS, hydrogen, accommodation and construction support as fields and mandates evolve.

> The window is the standardisation itself: the first operator to lease a validated, classed host — rather than fund another one from a blank sheet — resets the cost and schedule benchmark for the whole basin.

Leaseable offshore infrastructure — not a one-off build.

Offshore, the host hull is effectively long-lead equipment. The MPSS converts it from a custom project item into available leased infrastructure. A premium lease rate can be paid for by accelerated revenue alone — if first oil is pulled forward even by months.

FLEXIBLE COMMERCIAL STRUCTURES

- Time-charter & bareboat lease
- Joint venture & risk-share
- Lease-to-own and outright sale
- Lease-after-lease redeployment

A PREMIUM RATE, JUSTIFIED BY MORE THAN CAPEX

- Earlier first production & reduced FEED uncertainty
- Reduced fabrication and riser / cable-fatigue risk
- Easier topside expansion & field-life extension
- Redeployable asset value at the end of a charter

Because the hull is standardised and redeployable, a single asset can earn across multiple fields and missions over its life. That is the engine behind a durable, compounding platform rather than a single project return.

The shape of a single reference host.

Placeholder figures for discussion — the economics of one MPSS host leased on a stabilised charter. They are not an offer, quote or projection; every input is editable in the live returns model.

$600M	~15%	2x+	60%
Indicative all-in asset cost per host	Illustrative unlevered net yield on asset	Target equity MOIC over a ~10-year hold	Illustrative redeployable residual at exit

WHAT INVESTORS SCREEN FOR	THE MPSS ANSWER
Capital preservation	Hard-asset collateral — a classed steel hull with independent, redeployable residual value
Real, current income	Contracted multi-year day-rate charters — cash yield from year one of hire
Low correlation	Returns tied to offshore field economics and steel availability, not equity multiples or rate cycles
Inflation resilience	A physical, replacement-cost asset; charters can be structured with escalation
Exit optionality	Re-lease, sale-leaseback, lease-to-own, or outright sale of a redeployable asset

The figures above are illustrative placeholders for discussion only — not offers, projections, quotes or guarantees of return. See the full disclaimer at the foot of this memorandum.

INDEPENDENTLY MODELLED · 1989

An independent study found the MPSS improved investor returns on every field it modelled.

AUPEC — the Aberdeen University Petroleum & Economic Consultants, under its director **Prof. Alexander G. Kemp** (Professor of Economics, University of Aberdeen, and specialist adviser to the UK House of Commons Select Committee on Energy) — modelled five UK Continental Shelf fields under a conventional host versus the Seaways MPSS. In *every* case, post-tax returns to investors improved, driven by the MPSS's lower platform-structure cost — and marginal fields were turned into attractive investments.

MODEL UKCS FIELD	RECOVERABLE RESERVES	CONVENTIONAL HOST — POST-TAX IRR	SEAWAYS MPSS — POST-TAX IRR
Alba	250 mmbbl	21.3%	**25.5%**
Andrew	90 mmbbl · 200 bcf	11.6%	**16.9%**
Hudson	50 mmbbl	12.3%	**19.2%**
Keith	59 mmbbl	9.6%	**14.2%**
Machar	65 mmbbl	10.1%	**15.8%**

Source: AUPEC, "Market Potential and Economic Impact of the MPSS Production Vessel in the North Sea," Table S.3, November 1989. Post-tax returns at $20/bbl (constant real), 1989 prices. On the two most marginal fields the transformation was starkest — Keith moved from a negative NPV_{10} of −£3m to +£29m, and Machar from +£0.4m to +£37m (1989 prices). The cost advantage AUPEC modelled was borne out in the yard: in January 1990 the Clyde shipbuilder **Scott Lithgow** quoted about **£31.7m** for the base hull — 14,340 tonnes of steel, roughly **£2,200 per tonne** — for Texaco's Strathspey field (about £35.8m with moorings, roughly $57m at the time), well below typical semi-submersible build costs of the era. Presented as independent, historic validation of the MPSS's structural cost advantage — not a current projection, quote or offer.

Four decades of engineering, independently endorsed.

The MPSS was conceived by Craig Lang — founder of Seaways Engineering — in the early 1980s, drawing on deepwater tension-leg-platform mooring engineering. By the time of its development and independent review in Glasgow in 1988–89, much of the design was already complete. Since then its principles have been analysed, tank-tested, classed and published in the peer-reviewed literature — so a founding investor underwrites execution, not unproven science.

INDEPENDENT ENDORSEMENT

- Classification review by Lloyd's Register, ABS & DNV
- Reviewed under Prof. Douglas Faulkner, Head of Naval Architecture & Ocean Engineering, University of Glasgow
- Peer-reviewed: Faulkner, Incecik & Lang, "Design Evaluation of a Multi-Purpose Semi-Submersible" (read at the IESIS, 10 January 1989)
- Lloyd's Register pre-qualification certificate arranged; stability compliant with the Norwegian Maritime Directorate (1989)

TESTED & ANALYSED

- 1:100-scale model tested over 400 runs in the University of Glasgow towing tank (Incecik, report NAOE-89-01, 1989)
- **Designed for a service life beyond a century**; the conservative W S Atkins fatigue analysis set the documented minimum at 58 years
- Flooded (damaged) model kept its deck dry through storm seas of H_s in excess of 20 m
- Structural model tested at Glasgow (Conoco-sponsored); hydrodynamic tests sponsored by Texas Eastern North Sea

The documentary record

The MPSS is backed by a continuous, dated engineering and commercial record spanning five decades. Selected entries below; the full set is available in the data room. Documents are listed for provenance — their contents govern.

1975	**Reel-ship pipe-lay patent**, Craig Lang (US RE30,846) — the proven-technology lineage
1979	**Prof. Douglas Faulkner** compares the Hutton TLP on steel weight vs displacement
1986	**Seaways Engineering** — first development-funding document
1987	"A cheaper route to floating production" — **The Oilman** magazine
1988	Final report to **Texas Eastern North Sea** on the Phase 1 development programme
1988	**Lloyd's Register** principal-surveyor design review (G. Alford)
1988	**W S Atkins** Oil & Gas — interim engineering report
1988	**MPSS registered design**; SWIS storage bath test
1989	Peer-reviewed **IESIS** paper (Faulkner, Incecik & Lang); **Glasgow** 1:100 model testing (Incecik, NAOE-89-01)
1989	Structural report — **Prof. Faulkner** with the **University of Warwick** (strain-gauged)
1989	Independent economic study — **AUPEC / Prof. Alex Kemp**, University of Aberdeen
1990	**Scott Lithgow** (Clyde) shipyard quote to build the MPSS for Texaco's Strathspey field
1995	**Production-fluid storage patent** (GB2296686A) — the SWIS storage lineage

Entries reference documents held in the project archive and are provided for provenance only; the documents themselves are the authority and are available for review under the data room.

What founding capital builds.

The founding round capitalises **Lang Leasing** to carry the Seaways MPSS design across the last, decisive gap: from a class-endorsed drawing to a shipyard-ready, charter-backed first host — and the leasing company that owns it. Allocations below are illustrative and for discussion.

Engineering & class package	**35%**
Commercial & charter origination	**25%**
Shipyard & supply-chain readiness	**20%**
Platform, SPV & legal formation	**12%**
Working capital & contingency	**8%**

Path to a contracted first host

PHASE 1

Platform & package

Stand up the leasing entity; take the reference host to a shipyard-ready engineering and class package.

PHASE 2

Anchor charter

Originate a lead operator and structure the first multi-year day-rate charter against a defined field.

PHASE 3

Build & finance

Close asset-level financing into a ring-fenced SPV; commence standardised hull fabrication (~14 months).

PHASE 4

On hire & repeat

First host on charter and earning; the platform repeats the standardised build for the next field.

Back the standardisation of the offshore host.

We are inviting a small group of founding investors and strategic partners to capitalise Lang Leasing and the first host. The amount is an indicative founding-round range and the instrument is illustrative — the actual terms are shaped with founding investors in the data room.

Founding round

Founding equity to fund the first standardised host to a shipyard-ready, charter-backed position and to stand up Lang Leasing behind it.

$4–6M indicative range

Instrument	**Founding equity**
Use of proceeds	**Package · charter · platform**
Sponsor alignment	**Co-investment**
Structure	**Ring-fenced SPV per host**
Next step	**Data-room access**

Stewart Lang

Lang Leasing · founding-round lead

MPSS design: Craig Lang, founder, Seaways Engineering

stewart.lang@seaways-mpss.com

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All figures — including asset cost, yield, MOIC, hold period, residual value, use-of-proceeds allocations and the indicative founding-round range — are illustrative for discussion. They are not offers, projections, quotes, valuations or guarantees of return, and actual results may differ materially. Modelled or past performance does not indicate future results. Specifications, classification status, timelines and heritage are drawn from Seaways Engineering source material and remain subject to verification and independent diligence.

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